SCIENTIFIC CALCULATOR

A PYTHON TKINTER PROJECT

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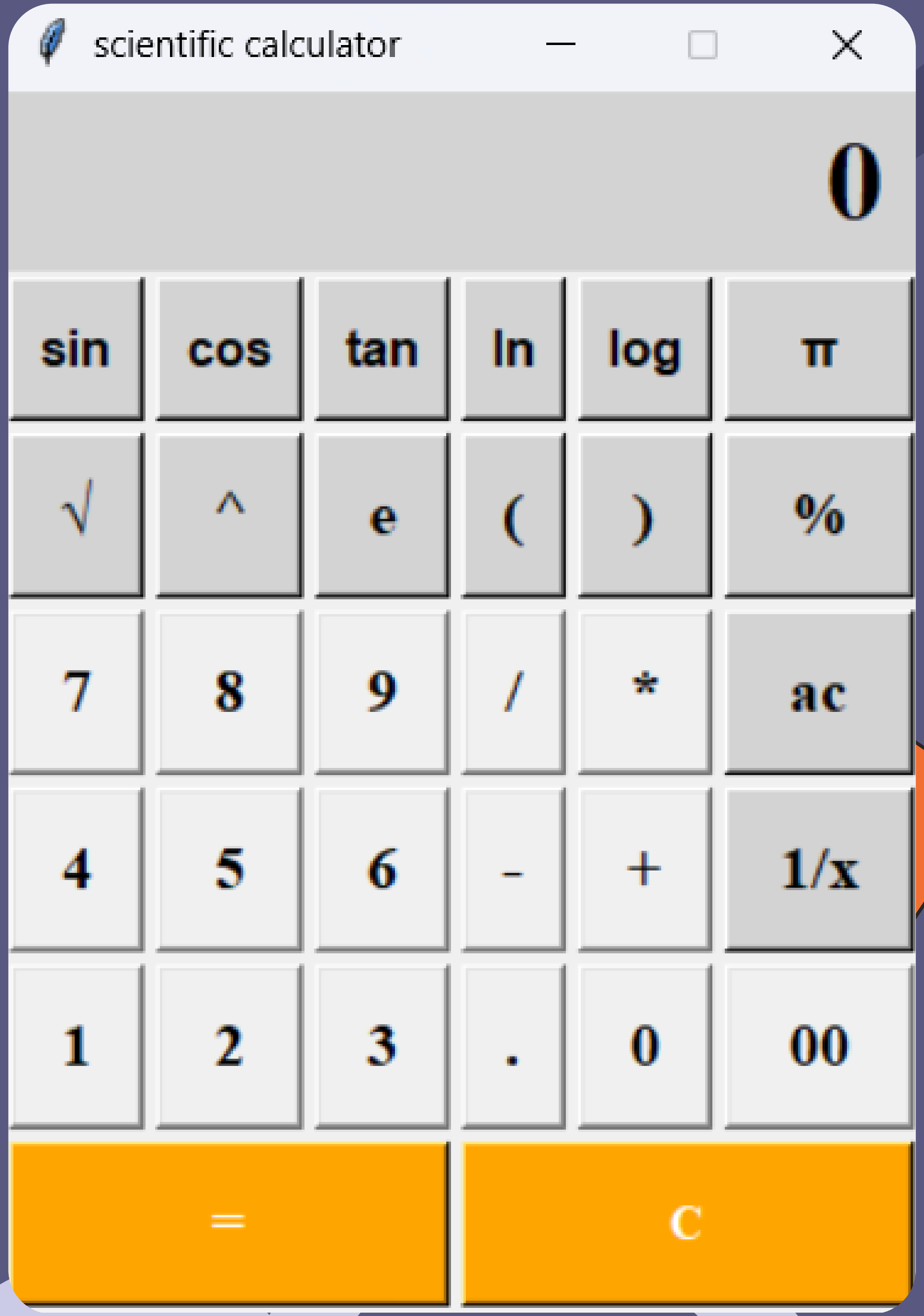
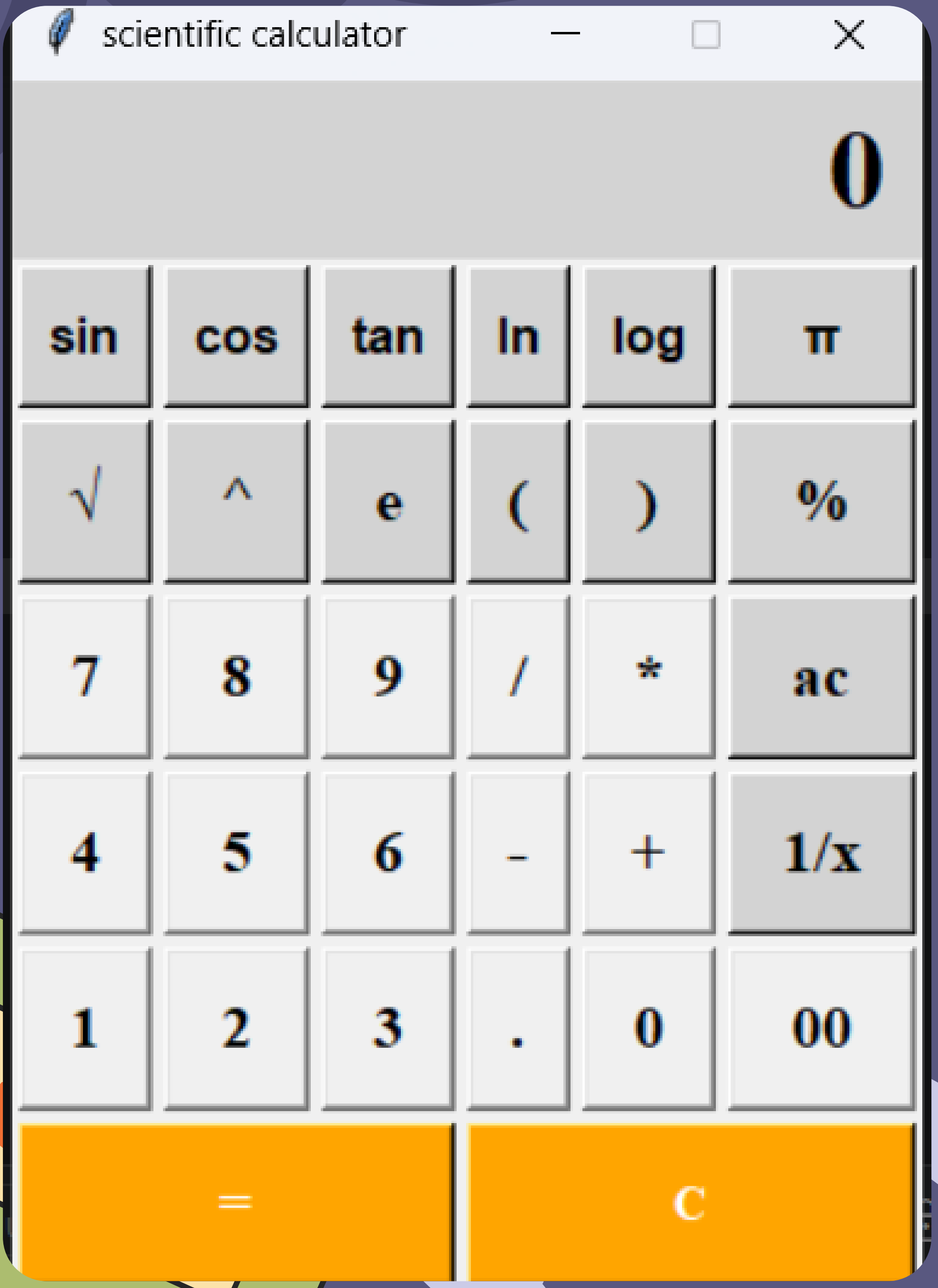
Introduction

- This project started as a basic calculator assignment.
- I upgraded it into a scientific calculator to make it more useful for real-world math.

- I chose Tkinter because:
  - It is Python's standard GUI library and easy for beginners.
  - Good documentation and simple layout options (grid, Button, Entry).





Features

- Basic operations:
  - Addition (+), subtraction (−), multiplication (×), division (÷), equals (=), clear, etc.
- Scientific functions I implemented:
  - Trigonometric: sin, cos, tan (working on angles in degrees).
  - Constants: π (pi) for circular calculations.
  - Logarithmic: log (base 10) and ln (natural log).
  - Roots and powers:
    - √ (square root)
    - ^ (power / exponentiation).

Challenges and Solutions

While building this calculator, I faced some problems:

- Too many buttons made the layout look messy.
- Wrong inputs or mistakes (like 1/0 or typing text instead of numbers) could crash the app.
- The calculator only worked with mouse clicks; I wanted it to work with keyboard too.

To solve these, I used Tkinter documentation and online resources and then:

- Neat layout:
  - I arranged all buttons in a grid (like rows and columns) so they look clean and organized.
  - I kept basic buttons (like +, -, ×, ÷) in one area and scientific buttons (like sin, cos, log, √) in another.

Conclusion

- This project started as a basic calculator but was upgraded into a scientific calculator using Python + Tkinter.
- The calculator supports daily math tasks as well as scientific operations like sin, cos, tan, log, ln, √, ^, and π.
- The interface is user-friendly, with a neat layout and support for both mouse clicks and keyboard inputs.
- Through this project, I improved my understanding of Tkinter GUI design, event handling, and error handling in Python applications.